Quantum Computing and Quantum Chemistry

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> State: very rough first draft.

1 Introduction

Chemistry occupies a prominent place in the list of potential applications of quantum computing. Particularly promising is the simulation of the quantum-mechanical aspects of chemistry: the structure and behaviour of molecules. And the top applications of this kind of simulation would be better drug design and new materials. But is this a realistic future? I have my doubts.

Here, though, are some typical mentions, from general audience public writing and from scientific research papers:

- "[A]ccurate quantum chemical calculations represent one of the most anticipated practical applications of quantum computers" (Caesura et al. 2025)

- “Accurate electronic structure calculations are considered one of the most anticipated applications of quantum computing that will revolutionize theoretical chemistry and other related fields.” (Arute et al. 2020)
- “Quantum computers are ideal for getting precise simulations of how potential drugs interact with complex biological molecules. This may help researchers better identify drug candidates – and ultimately improve healthcare worldwide.” (Google Quantum AI n.d.)
- “Puzzles like drug discovery or chemical synthesis could be rapidly accelerated by prototyping on a quantum computer. Quantum simulation would allow scientists to model complex molecules and simulate a drug’s reaction inside the human body, or run advanced test on chemicals without risk of harm or waste.” (Hazan et al., n.d.)
- “A quantum computer’s inherently quantum mechanical behavior can provide the exact solution to the Schrödinger equation mapped onto the qubits, without resorting to approximate methods.” (Andersson et al. 2022)
- “One of the most promising suggested applications of quantum computing is solving classically intractable chemistry problems.” (McArdle et al. 2020)
- “Chemical simulation is one of the most promising applications for future quantum computers. It is thought that quantum computers may enable accurate simulation for complex molecules that are otherwise impossible to simulate classically” (Goings et al. 2022)

Others I might want to use

Currently, these claims are made on the basis of plausibility arguments and extrapolations, at various levels of detail and specificity. So, as an ex-chemist, I wondered, do they hold water?

2 Feynman

In a 1982 talk speculating on the possibility of quantum computers, Richard Feynman exclaimed that “[N]ature isn’t classical, dammit, and if you want to make a simulation of nature, you’d better make it quantum mechanical.” (Feynman, n.d.) This talk has been cited almost 15,000 times. It is, in some sense, the founding statement of quantum computing

Still, like many good quotations, we should not look at it too closely or take it too literally. The design and potential usefulness of actual quantum computers did not come into focus for another decade. Feynman’s vision of quantum computing was not what has come to dominate the current mainstream: there was no mention of quantum circuits, gates, qubits, or entanglement. And, of course, few people argue that, just because optimization problems like packing problems or shortest-path problems are obviously classical, they are off limits to quantum computers. So, for all of his genius, let's put Mr. Feynman to one side.

3 Storage

Let's move on to the general landscape of today's quantum computers and quantum algorithms, with their circuits and operators and qubits. The general plausibility argument for quantum advantage goes something like this:

Let's imagine an approximate calculation of a molecule's electronic structure that gives a set of molecular orbitals ϕ_i. In this approximate calculation, the lowest energy orbitals will be occupied and the higher energy unoccupied. So let's represent that approximate state like this: $|11110000\rangle$.

But the exact state is (within the bounds of this set of molecular orbitals) a combination of each and every permutation of states with different orbitals occupied. That is, the number of terms in the following equation grows exponentially with the number of orbitals and electrons

$$|\Psi\rangle = \Sigma_i a_i |\Phi_0\rangle \quad (1)$$

Example:

For two electrons in a four-orbital model:

$$|\Psi\rangle = a_1|1100 > +a_2|1010\rangle + a_3|1001\rangle + a_4|0110\rangle + a_5|0101\rangle + a_6|0011\rangle \quad (2)$$

For a quantum computer, however, we can set up one qubit per orbital (so four cubits in this case) and by suitably mixing them up using operators, cover all the possibilities of equation Equation 2. Four is not a lot less than six, but as the number of orbitals and electrons grows, the number of terms that a classical computer needs to store grows exponentially, while the number of qubits is just the number of orbitals.

Here is how various publications express this advantage:

- On classical computers, resource requirements for complete simulation of the time-independent Schrödinger equation scale exponentially with the number of atoms in a molecule, limiting such full configuration interaction (FCI) calculations to diatomic and triatomic molecules… The calculation time for the energy of atoms and molecules scales exponentially with system size on a classical computer but polynomially using quantum algorithms. (aspuru-guzikSimulatedQuantumComputation2005?)
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